

Accounting and Accountability

Week 8



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Key concepts

Cost Classifications for Predicting Cost Behavior

Cost behavior refers to how a cost will react to changes in the level of activity.

The most common classifications are:

- **Variable costs.**
- **Fixed costs.**
- **Mixed costs.**

Key concepts

Variable Cost

A cost that varies, in total, in direct proportion to changes in the level of activity.

A variable cost per unit is constant.

Key concepts

Fixed Cost

A cost that remains constant, in total, regardless of changes in the level of the activity.

If expressed on a per unit basis, the average fixed cost per unit varies inversely with changes in activity.

Key concepts

Types of Fixed Costs

Committed

Long-term, cannot be significantly reduced in the short term

Discretionary

May be altered in the short-term by current managerial decisions

Key concepts

Behavior of Cost (within the relevant range)		
Cost	In Total	Per Unit
Variable	Total variable cost Increase and decrease in proportion to changes in the activity level.	Variable cost per unit remains constant.
Fixed	Total fixed cost is not affected by changes in the activity level within the relevant range.	Fixed cost per unit decreases as the activity level rises and increases as the activity level falls.

The **relevant range** is the anticipated production activity level. Fixed costs remain constant within a relevant range. If production levels exceed expectations, then additional fixed costs will be required.

Quick Check

Which of the following costs would be variable with respect to the number of ice cream cones sold at a Baskin & Robbins? (There may be more than one correct answer.)

- A. The cost of lighting the store.
- B. The wages of the store manager.
- C. The cost of ice cream.
- D. The cost of napkins for customers.

Quick Check

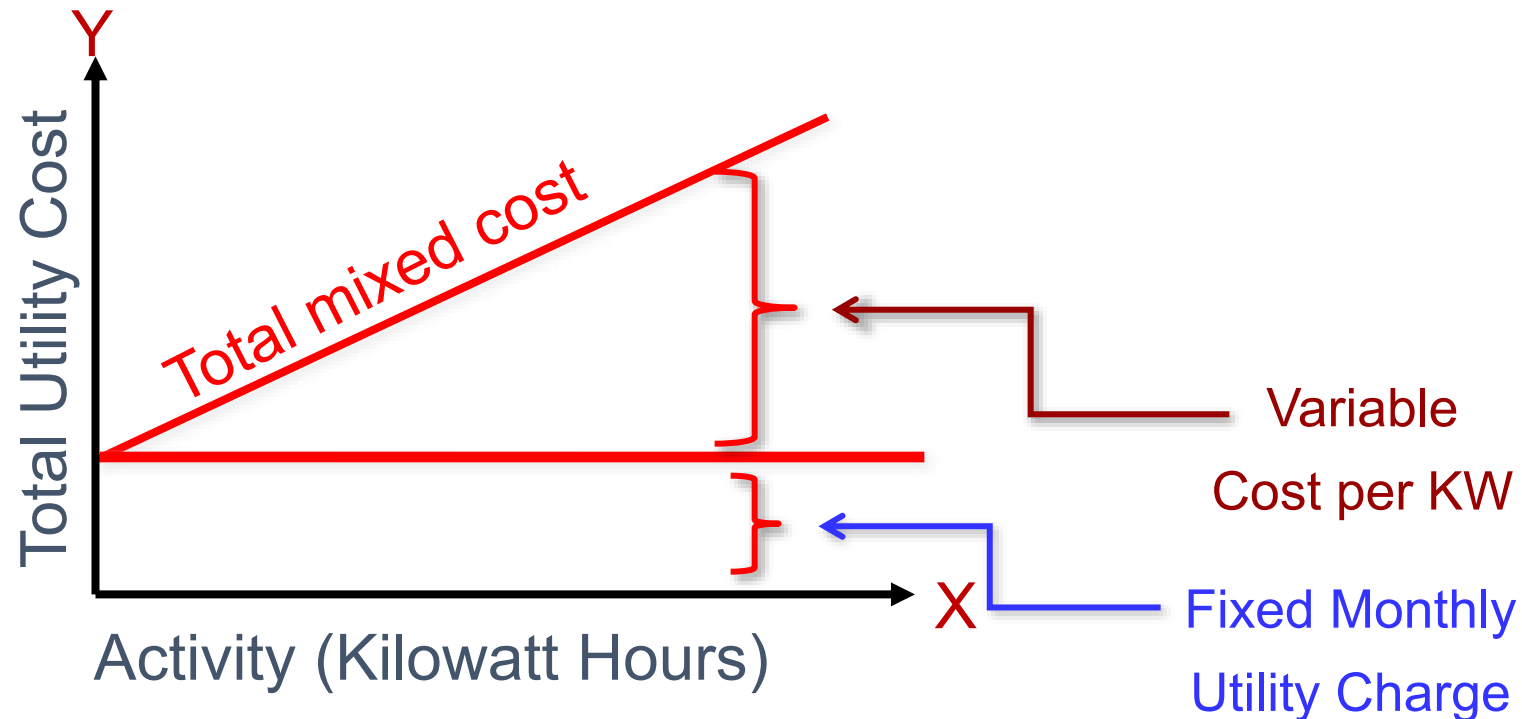
Which of the following costs would be variable with respect to the number of ice cream cones sold at a Baskin & Robbins? (There may be more than one correct answer.)

- A. The cost of lighting the store.
- B. The wages of the store manager.
- ☒ C. The cost of ice cream.
- ☒ D. The cost of napkins for customers.

Key concepts

Mixed Costs – Part 1

A mixed cost contains both variable and fixed elements.
Consider the example of utility cost.



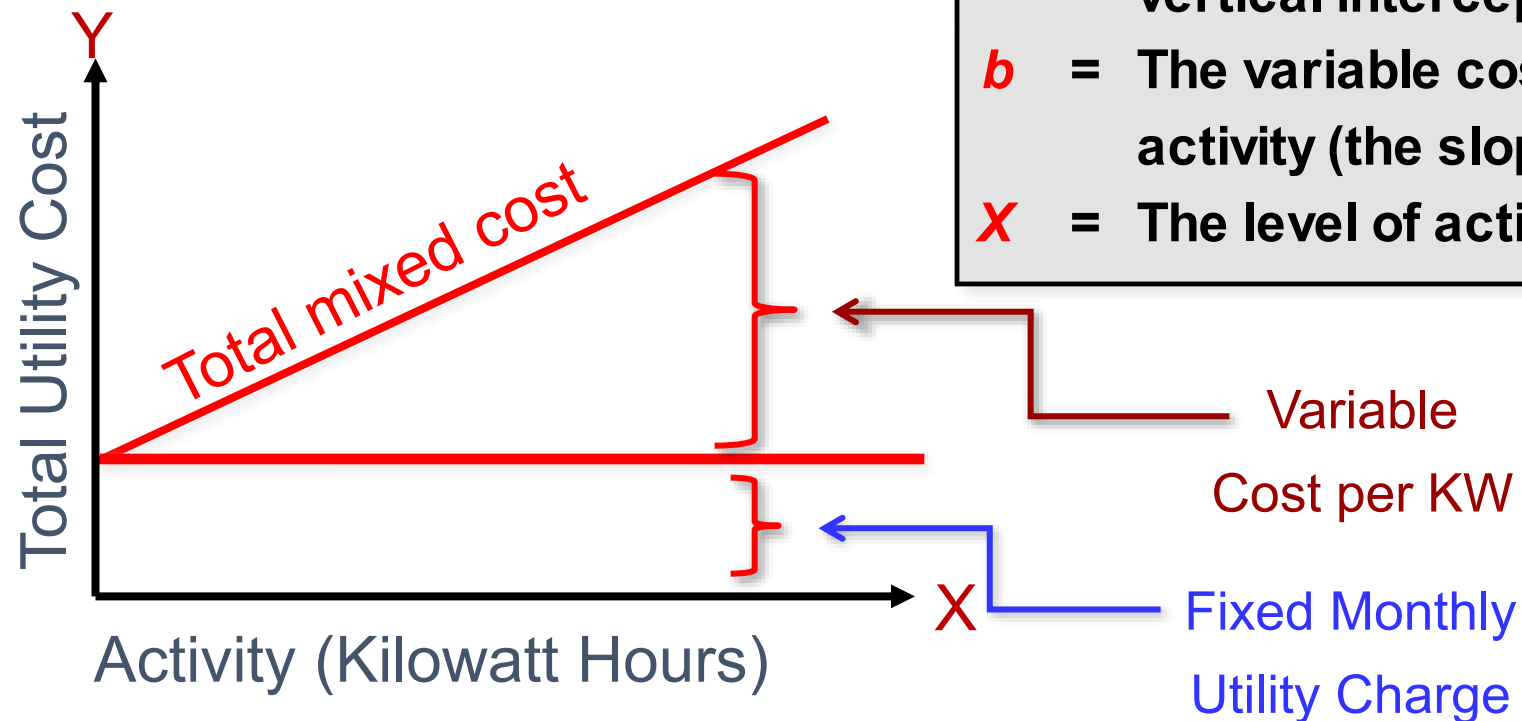
Key concepts

Mixed Costs – Part 1

The total mixed cost line can be expressed as an equation: $Y = a + bX$

Where:

- Y = The total mixed cost.
- a = The total fixed cost (the vertical intercept of the line).
- b = The variable cost per unit of activity (the slope of the line).
- X = The level of activity.



Key concepts

Mixed Costs – An Example

If your fixed monthly utility charge is \$40, your variable cost is \$0.03 per kilowatt hour, and your monthly activity level is 2,000 kilowatt hours, what is the amount of your utility bill?

$$Y = a + bX$$

$$Y = \$40 + (\$0.03 \times 2,000)$$

$$Y = \textbf{\$100}$$

Key concepts

Mixed Costs – Another Example

Let's take a more in-depth look at the cost equation by examining the costs incurred by Amantha's Artistry in the manufacture of sweet treats, as shown in the table below.

Cost Information for Amantha's Artistry

Cost Incurred	Fixed or Variable	Cost
Rent on premises	Fixed	\$20,000 per year
Public liability insurance	Fixed	\$15,000 per year
Ingredients	Variable	\$2 per treat
Staff labour	Variable	\$1 per treat

By applying the cost equation, Amantha's Artistry can predict its costs at any level of activity (x) as follows:

1. Determine total fixed costs: $\$20,000 + \$15,000 = \$35,000$
2. Determine variable costs per unit: $\$2 + \$1 = \$3$
3. Complete the cost equation: $Y = \$35,000 + \$3x$

Key concepts

The Traditional and Contribution Formats

Comparison of the Contribution Income Statement with the Traditional Income Statement			
Traditional Format		Contribution Format	
Sales	\$ 100,000	Sales	\$ 100,000
Cost of goods sold	70,000	Variable expenses	60,000
Gross margin	\$ 30,000	Contribution margin	\$ 40,000
Selling & admin. expense	20,000	Fixed expenses	30,000
Net operating income	\$ 10,000	Net operating income	\$ 10,000

Used primarily for
external reporting.

Used primarily by
management.

Key concepts

Why is CVP analysis?

- Balancing Cost, Volume, and Profit. Managers employ cost-volume-profit (CVP) analysis to determine the sales level at which they break even or balance their revenue with their expenses.



Key concepts

Contribution Margin (CM)

The contribution income statement is helpful to managers in judging the impact on profits of changes in selling price, cost, or volume. The emphasis is on cost behavior.

Racing Bicycle Company Contribution Income Statement For the Month of June		
Sales (500 bicycles)	\$	250,000
Less: Variable expenses		150,000
Contribution margin		100,000
Less: Fixed expenses		80,000
Net operating income	\$	20,000

Contribution Margin (CM) is the amount remaining from sales revenue after variable expenses have been deducted.

Key concepts

Contribution Margin (CM)

CM is used first to cover fixed expenses.
Any remaining CM contributes to net operating income.

Racing Bicycle Company Contribution Income Statement For the Month of June		
Sales (500 bicycles)	\$	250,000
Less: Variable expenses		150,000
Contribution margin		100,000
Less: Fixed expenses		80,000
Net operating income	\$	20,000

Key concepts

Contribution Margin per unit

Sales, variable expenses, and contribution margin can also be expressed on a per unit basis. If Racing sells an additional bicycle, **\$200** additional CM will be generated to cover fixed expenses and profit.

Racing Bicycle Company Contribution Income Statement For the Month of June		
	Total	Per Unit
Sales (500 bicycles)	\$ 250,000	\$ 500
Less: Variable expenses	150,000	300
Contribution margin	100,000	\$ 200
Less: Fixed expenses	80,000	
Net operating income	<u>\$ 20,000</u>	

Key concepts

Contribution Margin ratio

$$\text{Contribution Margin Ratio} = \frac{\text{Contribution Margin per Unit}}{\text{Sales Price per Unit}}$$

Racing Bicycle Company Contribution Income Statement For the Month of June		
	Total	Per Unit
Sales (500 bicycles)	\$ 250,000	\$ 500
Less: Variable expenses	150,000	300
Contribution margin	100,000	\$ 200
Less: Fixed expenses	80,000	
Net operating income	\$ 20,000	

$$\text{CMR} = 200 / 500 = 40\%$$

$$\text{Total Sales} \times \text{Contribution Margin Ratio} = \text{Total Contribution Margin}$$

Key concepts

- **Assumptions required for cost-volume-profit analysis**
- 1. Costs are linear and can clearly be designated as either fixed or variable.
- 2. Selling price per unit remains constant and does not increase or decrease based on volume .
- 3. In the case of manufacturing businesses, inventory does not change because we make the assumption that all units produced are sold.
- 4. In the case of a company that sells multiple products, the sales mix remains constant.

Key concepts

Break-even point

Each month, RBC must generate at least \$80,000 in total contribution margin to break-even (which is the level of sales at which profit is zero). If RBC sells **400 units** in a month, it will be operating at the *break-even point*.

Racing Bicycle Company Contribution Income Statement For the Month of June		
	Total	Per Unit
Sales (400 bicycles)	\$ 200,000	\$ 500
Less: Variable expenses	120,000	300
Contribution margin	80,000	\$ 200
Less: Fixed expenses	80,000	
Net operating income	\$ -	

Key concepts

Break-even point

$$\text{Break-Even Point in Units: } \frac{\text{Total Fixed Costs}}{\text{Contribution Margin per Unit}}$$

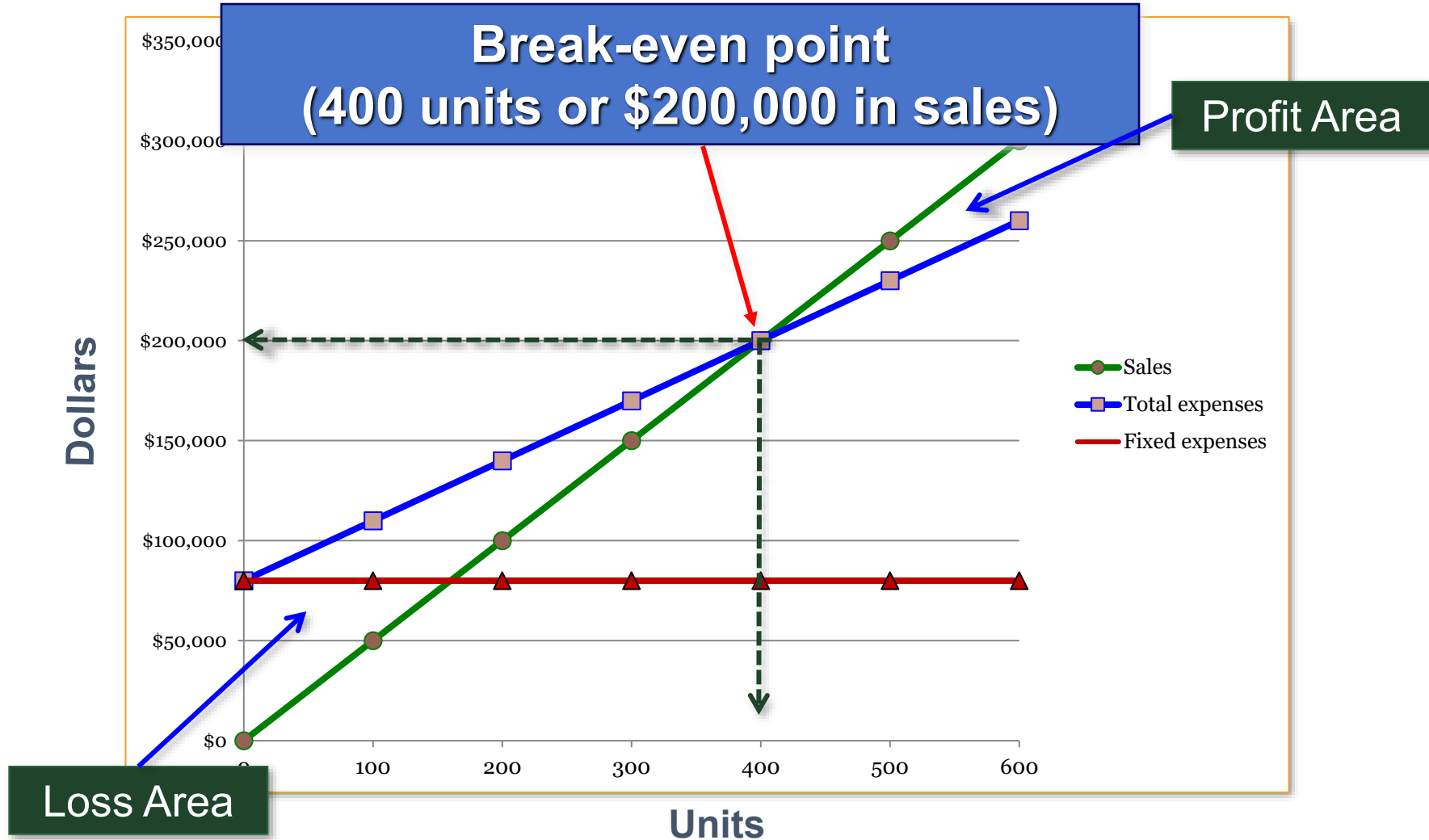
$$\text{Break-Even Point in Dollars} = \frac{\text{Fixed Costs}}{\text{Contribution Margin Ratio}}$$

Racing Bicycle Company Contribution Income Statement For the Month of June

	Total	Per Unit
Sales (400 bicycles)	\$ 200,000	\$ 500
Less: Variable expenses	120,000	300
Contribution margin	80,000	\$ 200
Less: Fixed expenses	80,000	
Net operating income	\$ -	

Key concepts

Break-Even Point



Key concepts

CVP Relationships in Equation Form

The contribution format income statement can be expressed in the following equation:

$$\text{Profit} = (\text{Sales} - \text{Variable expenses}) - \text{Fixed expenses}$$

Racing Bicycle Company Contribution Income Statement For the Month of June		
	Total	Per Unit
Sales (401 bicycles)	\$ 200,500	\$ 500
Less: Variable expenses	120,300	300
Contribution margin	80,200	\$ 200
Less: Fixed expenses	80,000	
Net operating income	\$ 200	

Key concepts

CVP Relationships in Equation Form – Example

This equation can be used to show the profit RBC earns if it sells 401. Notice, the answer of \$200 mirrors our earlier solution.

$$\text{Profit} = (\text{Sales} - \text{Variable expenses}) - \text{Fixed expenses}$$

$$401 \text{ units} \times \$500$$

$$401 \text{ units} \times \$300$$

$$\$80,000$$

$$\text{Profit} = (\$200,500 - \$120,300) - \$80,000$$

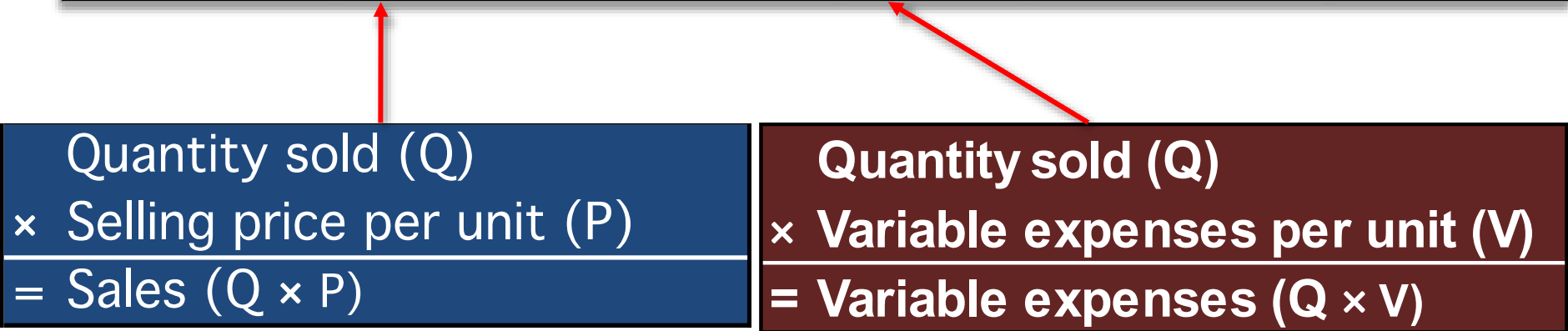
$$\text{\textcolor{red}{\$200}} = (\$200,500 - \$120,300) - \$80,000$$

Key concepts

CVP Relationships in Equation Form – Detail Breakdown

When a company has only one product we can further refine this equation as shown on this slide.

$$\text{Profit} = (\text{Sales} - \text{Variable expenses}) - \text{Fixed expenses}$$



Quantity sold (Q)
× Selling price per unit (P)
= Sales (Q × P)

Quantity sold (Q)
× Variable expenses per unit (V)
= Variable expenses (Q × V)

$$\text{Profit} = (P \times Q - V \times Q) - \text{Fixed expenses}$$

Key concepts

CVP Relationships in Equation Form – Example Showing Detail

This equation can also be used to show the \$200 profit RBC earns if it sells 401 bikes.

$$\text{Profit} = (\text{Sales} - \text{Variable expenses}) - \text{Fixed expenses}$$

$$\text{Profit} = (P \times Q - V \times Q) - \text{Fixed expenses}$$

$$\$200 = (\$500 \times 401 - \$300 \times 401) - \$80,000$$

Key concepts

CVP Relationships in Equation Form – Using Unit Contribution Margin

It is often useful to express the simple profit equation in terms of the unit contribution margin (Unit CM) as follows. This equation can also be used to compute RBC's \$200 profit if it sells 401 bikes.

$$\begin{aligned}\text{Profit} &= (P \times Q - V \times Q) - \text{Fixed expenses} \\ \text{Profit} &= (P - V) \times Q - \text{Fixed expenses} \\ \text{Profit} &= \text{Unit CM} \times Q - \text{Fixed expenses}\end{aligned}$$

$$\text{Profit} = (\$500 - \$300) \times 401 - \$80,000$$

$$\text{Profit} = \$200 \times 401 - \$80,000$$

$$\text{Profit} = \$80,200 - \$80,000$$

$$\text{Profit} = \$200$$

Unit CM = Selling price per unit – Variable expenses per unit

Key concepts

CVP Relationships in Equation Form – Using Contribution Margin ratio

It is often useful to express the simple profit equation in terms of the contribution margin ratio as follows. This equation can also be used to compute RBC's \$200 profit if it sells 401 bikes.

$$\text{Profit} = (\text{CM ratio} \times \text{Sales}) - \text{Fixed expenses}$$

$$\text{Profit} = 40\% \times 200,500 - \$80,000$$

$$\text{Profit} = \$80,200 - \$80,000$$

$$\text{Profit} = \$200$$

Quick Check

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. An average of 2,100 cups are sold each month. What is the break-even sales dollars?

- a. \$1,300
- b. \$1,715
- c. \$1,788
- d. \$3,129

Quick Check

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. An average of 2,100 cups are sold each month. What is the break-even sales dollars?

- a. \$1,300
- ☒ b. \$1,715
- c. \$1,788
- d. \$3,129

$$\begin{aligned}\text{Break-even sales} &= \frac{\text{Fixed expenses}}{\text{CM Ratio}} \\ &= \frac{\$1,300}{0.758} \\ &= \$1,715\end{aligned}$$

Quick Check

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. An average of 2,100 cups are sold each month. What is the break-even sales in units?

- a. 872 cups
- b. 3,611 cups
- c. 1,200 cups
- d. 1,150 cups

Quick Check

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. An average of 2,100 cups are sold each month. What is the break-even sales in cups?

- a. 872 cups
- b. 3,611 cups
- c. 1,200 cups
- d. 1,150 cups

$$\begin{aligned}\text{Break-even} &= \frac{\text{Fixed expenses}}{\text{CM per Unit}} \\ &= \frac{\$1,300}{\$1.49/\text{cup} - \$0.36/\text{cup}} \\ &= \frac{\$1,300}{\$1.13/\text{cup}} \\ &= 1,150 \text{ cups}\end{aligned}$$

Quick Check

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. An average of 2,100 cups are sold each month. What is the CM Ratio for Coffee Klatch?

- a. 1.319
- b. 0.758
- c. 0.242
- d. 4.139

Quick Check

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. An average of 2,100 cups are sold each month. What is the CM Ratio for Coffee Klatch?

- a. 1.319
- ☒ b. 0.758
- c. 0.242
- d. 4.139

$$\begin{aligned}\text{CM Ratio} &= \frac{\text{Unit contribution margin}}{\text{Unit selling price}} \\ &= \frac{(\$1.49 - \$0.36)}{\$1.49} \\ &= \frac{\$1.13}{\$1.49} = 0.758\end{aligned}$$

Key concepts

Target Profit Analysis--Unit sales

Suppose RBC wants to know how many bikes must be sold to earn a profit of \$100,000.

$$\text{Unit sales to attain the target profit} = \frac{\text{Target profit} + \text{Fixed expenses}}{\text{CM per unit}}$$

$$\text{Unit sales} = \frac{\$100,000 + \$80,000}{\$200}$$

$$\text{Unit sales} = 900 \text{ units}$$

Key concepts

Target Profit Analysis--Dollar sales

$$\text{Dollar sales to attain the target profit} = \frac{\text{Target profit} + \text{Fixed expenses}}{\text{CM ratio}}$$

$$\text{Dollar sales} = \frac{\$100,000 + \$80,000}{40\%}$$

$$\text{Dollar sales} = \$450,000$$

Quick Check

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. Determine **how many cups** of coffee would have to be sold to attain target profits of \$2,500 per month.

- a. 3,363 cups
- b. 2,212 cups
- c. 1,150 cups
- d. 4,200 cups

Quick Check

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The

average fixed expense per cup is \$0.36.

How many cups of coffee must be sold to attain target profit?

Target profit is \$3,800.

a. 3,363 cups

b. 2,212 cups

c. 1,150 cups

d. 4,200 cups

Unit sales
to attain
target profit

$$= \frac{\text{Target profit} + \text{Fixed expenses}}{\text{Unit CM}}$$

$$= \frac{\$2,500 + \$1,300}{\$1.49 - \$0.36}$$

$$= \frac{\$3,800}{\$1.13}$$

$$= 3,363 \text{ cups}$$

Quick Check

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. Determine the **sales dollars** that must be generated to attain target profits of \$2,500 per month.

- a. \$2,550
- b. \$5,013
- c. \$8,458
- d. \$10,555

Quick Check

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The

average fixed

sales dollars

\$2,500 per month

a. \$2,550

b. \$5,013

c. \$8,458

d. \$10,555

$$\begin{aligned}\text{Sales \$ to attain target profit} &= \frac{\text{Target profit} + \text{Fixed expenses}}{\text{CM ratio}} \\ &= \frac{\$2,500 + \$1,300}{(\$1.49 - 0.36) \div \$1.49} \\ &= \frac{\$3,800}{0.758} \\ &= \$5,013\end{aligned}$$

Additional Applications of CVP Concepts – Example 1

Example 1: Change in Fixed Cost and Sales Volume

What is the profit impact if RBC can increase unit sales from 500 to 540 by increasing the monthly advertising budget by \$10,000?

Additional Applications of CVP Concepts – Example 1

Example 1: Change in Fixed Cost and Sales Volume

$\$80,000 + \$10,000 \text{ advertising} = \$90,000$

	500 units	540 units
Sales	\$ 250,000	\$ 270,000
Less: Variable expenses	150,000	162,000
Contribution margin	100,000	108,000
Less: Fixed expenses	80,000	90,000
Net operating income	\$ 20,000	\$ 18,000

Sales *increased* by \$20,000, but net operating income *decreased* by \$2,000.

Additional Applications of CVP Concepts – A Shortcut

Example 1: Change in Fixed Cost and Sales Volume

A shortcut solution using incremental analysis

Increase in CM (40 units X \$200)	\$ 8,000
Increase in advertising expenses	<u>10,000</u>
Decrease in net operating income	<u><u>\$ (2,000)</u></u>

Additional Applications of CVP Concepts – Example 2

Example 2: Change in Variable Costs and Sales Volume

What is the profit impact if RBC can use higher quality raw materials, thus increasing variable costs per unit by \$10, to generate an increase in unit sales from 500 to 580?

Additional Applications of CVP Concepts – Example 2

Example 2: Change in Variable Costs and Sales Volume

$$580 \text{ units} \times \$310 \text{ variable cost/unit} = \$179,800$$

	500 units	580 units
Sales	\$ 250,000	\$ 290,000
Less: Variable expenses	150,000	179,800
Contribution margin	100,000	110,200
Less: Fixed expenses	80,000	80,000
Net operating income	\$ 20,000	\$ 30,200

Sales *increase* by \$40,000 and net operating income *increases* by \$10,200.

Additional Applications of CVP Concepts – Example 3

Example 3: Change in Fixed Cost, Selling Price, and Sales Volume

What is the profit impact if RBC:

1. cuts its selling price \$20 per unit,
2. increases its advertising budget by \$15,000 per month, and
3. increases sales from 500 to 650 units per month?

Additional Applications of CVP Concepts – Example 3

Example 3: Change in Fixed Cost, Selling Price, and Sales Volume

$$650 \text{ units} \times \$480 = \$312,000$$

	500 units	650 units
Sales	\$ 250,000	\$ 312,000
Less: Variable expenses	150,000	195,000
Contribution margin	100,000	117,000
Less: Fixed expenses	80,000	95,000
Net operating income	\$ 20,000	\$ 22,000

Sales **increase** by \$62,000, fixed costs **increase** by \$15,000, and net operating income **increases** by \$2,000.

Additional Applications of CVP Concepts – Example 4

Example 4: Change in Variable Cost, Fixed Cost, and Sales Volume

What is the profit impact if RBC:

1. pays a \$15 sales commission per bike sold instead of paying salespersons flat salaries that currently total \$6,000 per month, and
2. increases unit sales from 500 to 575 bikes?

Additional Applications of CVP Concepts – Example 4

Example 4: Change in Variable Cost, Fixed Cost, and Sales Volume

$$575 \text{ units} \times \$315 = \$181,125$$

	500 units	575 units
Sales	\$ 250,000	\$ 287,500
Less: Variable expenses	150,000	181,125
Contribution margin	100,000	106,375
Less: Fixed expenses	80,000	74,000
Net operating income	<u>\$ 20,000</u>	<u>\$ 32,375</u>

Sales *increase* by \$37,500, fixed expenses *decrease* by \$6,000, and net operating income *increases* by \$12,375.

Additional Applications of CVP Concepts – Example 5

Example 5: Change in Selling Price

If RBC has an opportunity to sell 150 bikes to a wholesaler without disturbing sales to other customers or fixed expenses, what price would it quote to the wholesaler if it wants to increase monthly profits by \$3,000?

Additional Applications of CVP Concepts – Example 5

Example 5: Change in Selling Price

\$ 3,000 ÷ 150 bikes	=	\$ 20 per bike
Variable cost per bike	=	<u>300 per bike</u>
Selling price required	=	<u><u>\$ 320 per bike</u></u>

150 bikes × \$320 per bike	=	\$ 48,000
Total variable costs	=	<u>45,000</u>
Increase in net operating income	=	<u><u>\$ 3,000</u></u>

Key concepts-- break-even analysis in a multi-product environment

The Definition of Sales Mix

- Sales mix is the relative proportion in which a company's products are sold.
- Different products have different selling prices, cost structures, and contribution margins.
- When a company sells more than one product, break-even analysis becomes more complex as the following example illustrates.

Key concepts-- break-even analysis in a multi-product environment

Let's meet Soul Sisters – they are a social enterprise who work with refugee women to help develop their seamstress skills while producing ethical fashion. They currently make three clothing items – a long sleeve blouse, a pair of work pants and a work skirt with pockets. Each item has its own sales price, variable cost and contribution margin, as shown below:

	Sales price per item	Variable cost per item	Contribution margin
Long sleeve blouse	\$150	\$75	\$75
Work pants	\$140	\$50	\$90
Work skirt	\$115	\$65	\$50

Key concepts-- break-even analysis in a multi-product environment

The sales mix is that if the business sold 10 products – 6 would be blouses, 3 would be work pants and 1 would be a work skirt.

Our next step is to calculate a weighted average contribution margin:

	Contribution margin	Mix	Weighted contribution margin (CM x Mix)
Long sleeve blouse	\$75	60%	\$45
Work pants	\$90	30%	\$27
Work skirt	\$50	10%	\$5
Weighted average contribution margin for one unit			\$77

Key concepts-- break-even analysis in a multi-product environment

What is the break even point? To do this, we need to know the fixed costs. Soul Sisters tells us that fixed costs are \$13,860. We use the weighted average contribution margin in the same way as we would use the contribution margin to calculate the break even point:

$$\begin{aligned}\text{Break even}_{\text{units}} &= \text{Fixed costs} / \text{Weighted average contribution margin} \\ &= \$13,860 / \$77 \\ &= 180 \text{ units}\end{aligned}$$

Key concepts-- break-even analysis in a multi-product environment

However, we don't actually sell this weighted average unit. So we need to break up the 180 units into the 3 different products using our mix of 60%, 30% and 10%.

	Total break even units	Mix	Break even sales in units
Long sleeve blouse	180	60%	108
Work pants	180	30%	54
Work skirt	180	10%	18

Using a forecasted or estimated contribution margin income statement, let's verify that the break even sales in units at Soul Sisters is correct.

SOUL SISTERS Forecasted Contribution Margin Income Statement	
Sales	
Long sleeve blouse (108 x \$75)	16,200
Work pants (54 x \$90)	7,560
Work skirt (18 x \$50)	2,070
Total sales	25,830
Less Variable Costs	
Long sleeve blouse (108 x \$75)	8,100
Work pants (54 x \$50)	2,700
Work skirt (18 x \$65)	1,170
Total variable costs	11,970
CONTRIBUTION MARGIN	13,860
Fixed costs	13,860
NET PROFIT/INCOME	0

Tutorial activities--Tutorial 8A

Which is the correct answer for a fixed cost?

- a. Total fixed cost changes as units change.
- b. Total fixed cost always stays the same.
- c. Fixed costs per unit increase as units increase.
- d. Fixed costs per unit always stays the same.

A business has one product with a sales price of \$40, variable cost of \$15 and annual fixed costs of \$50,000. They would like to earn a target profit after tax of \$40,000. The tax rate is 25%. What is the break-even point in units?

- a. 2,000
- b. 909
- c. 1,250
- d. 3,333

Tutorial activities--Tutorial 8A

A business has one product with a sales price of \$40, variable cost of \$15 and annual fixed costs of \$50,000. They would like to earn a target profit after tax of \$40,000. The tax rate is 25%. How many units need to be sold to earn the target profit after tax? Round your answer at the end of your calculation to the nearest unit (note: use normal maths rounding).

- a.2,000
- b.4,133
- c.3,600
- d.3,333

Following the above question, if the business increases the sales price by 20% what is the new break-even point? Round your answer to the nearest unit.

- a.7,143
- b.2,273
- c.3,600
- d.1,515

Tutorial activities--Tutorial 8A CVP analysis

Marketing Love ♥ makes and sells corporate gift baskets that clients can use for marketing campaigns and gifts.

<u>The cost to make each basket</u>			
Item	Cost per item	Quantity per basket	Total cost
Basket	\$3	1	\$3
Consumables	\$5	5	\$25
Branded gift	\$10	1	\$10
Packaging	\$4	1	\$4
Labour per basket	\$30 per hour	0.2 hours	\$6
Total basket cost			\$48

<u>Other business information</u>	
Sales price per basket	\$100
Business insurance per year	\$1,200
Monthly business labour costs (not related to making baskets)	\$3,000
Advertising per month	\$200
Factory rent and bills per month	\$4,000
Courier costs per basket	\$10
Tax rate	25%

Tutorial activities--Tutorial 8A CVP analysis

1. What is the variable cost per unit?
2. What is the total fixed cost for the year?
3. What is the contribution margin per unit?
4. What is the break-even point in units? Round your answer to the nearest unit. Analyse this result and the business information above and explain to Marketing Love what this number means.
5. What is the break-even point in dollars?
6. The business is currently selling 2,000 baskets. Prepare a contribution margin income statement.

Sales revenue	
- Variable costs	
= Contribution margin	
- Fixed costs	
= Profit (Loss) before tax	

7. How many baskets need to be sold to earn \$50,000 after tax? Round your answer right at the end of the calculation to the nearest basket. What does this answer mean?
8. Given all these calculations, what would you recommend changing to make this business more profitable? Come up with three different scenarios, where you change *one* thing, that might make this business more profitable. Give the scenario and show how it will change the break-even point.
9. Extension question: Can you come up with one alternative scenario, where you change a few factors at the same time to make this business more profitable. Give the scenario and show how it will change the break-even point.

Tutorial problems week 8

Follow the guide of questions in the previous Marketing Love ♥ CVP analysis, pick your interested product and do a similar CVP analysis and present your findings.

- You will work in groups (4-5 in one group) on tutorial problems.
- You will prepare a PPT and present the solution with the class in tutorial week.
- All group members need to participate and explain the solution and also answer any further questions from the class.
- Time limit: 15-20 minutes.

Tutorial activities--Tutorial 8B

Single and multiple product CVP analysis

SCENARIO 1

	Product: Wireless earbuds	
Sales price	\$50.00	
Variable cost	\$20.00	
Fixed costs	\$400.00	per month
Contribution margin per unit		
Break even (units)		raw calculation
		whole units
Target profit	\$2,000.00	<= Target profit per month before tax
Target sales (units)		raw result
		whole units (rounded)

Contribution margin profit and loss statement

	Break even	Target profit
Sales		
Less Variable Costs		
Contribution margin		
Fixed costs		
Profit before tax		
Less tax @30%		
Net profit/income after tax		

Tutorial activities--Tutorial 8B

Single and multiple product CVP analysis

SCENARIO 2		Increase the selling price by \$10	
		Product: Wireless earbuds	
Sales price			
Variable cost			
Fixed costs			
Contribution margin per unit			
Break even (units)			raw calculation
			whole units
Target profit		\$2,000.00	<= Target profit per month before tax
Target sales (units)			raw result
			whole units (rounded)

Contribution margin profit and loss statement

	Break even	Target profit
Sales		
Less Variable Costs		
Contribution margin		
Fixed costs		
Profit before tax		
Less tax @30%		
Net profit/income after tax		

Tutorial activities--Tutorial 8B Single and multiple product CVP analysis

SCENARIO 3		Decrease the fixed costs per month by \$50.	
		Product: Wireless earbuds	
Sales price			
Variable cost			
Fixed costs			
Contribution margin per unit			
Break even (units)		raw calculation	
		whole units	
Target profit	\$2,000.00	<= Target profit per month before tax	
Target sales (units)		raw result	
		whole units (rounded)	
Contribution margin profit and loss statement			
		Break even	Target profit
Sales			
Less Variable Costs			
Contribution margin			
Fixed costs			
Profit before tax			
Less tax @30%			
Net profit/income after tax			

Tutorial activities--Tutorial 8B Single and multiple product CVP analysis

SCENARIO 4		The business has 3 products		
		Earbuds	Headsets	Speakers
Sales price		\$50.00	\$70.00	\$99.00
Variable cost		\$20.00	\$30.00	\$42.00
Fixed costs		\$2,000.00 per month		
Contribution margin per unit				
Sales mix		50%	30%	20%
Weighted average contribution margin				
Break even (units) TOTAL			raw calculation	
			whole units	
Break even (units) per product raw calculation				
Break even (units) per product whole units				
Target profit		\$2,000.00	<= Target profit per month before tax	
Targetsales (units) TOTAL			raw result	
			whole units	
Target sales per product raw calculation				
Target sales per product whole units				

Tutorial activities--Tutorial 8B

Single and multiple product CVP analysis

Contribution margin profit and loss statement	Break even	Target profit
Sales		
Less Variable Costs		
Contribution margin		
Fixed costs		
Profit before tax		
Less tax @30%		
Net profit/income after tax		

- If the business had to focus only on one product, which one should they focus on?
- The owners of this business have asked for your advice about how they might increase the profitability of this business. What would you suggest based on the information in scenario 4?